Project Proposal

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Project Title

T2S: An AI-Powered Multi-Agent Intelligent Platform for Student Skill Development, Mentorship, Collaboration, and Career Discovery

1. Introduction

1.1 Background

In universities, students often have difficulty finding appropriate mentors, developing practical technical skills, discovering research opportunities, forming project teams, participating in hackathons, and identifying career opportunities that match their abilities.

Although universities may have teachers, student clubs, research groups, alumni networks, career services, and learning resources, these activities are often distributed across different platforms and communication channels.

For example, a student interested in Artificial Intelligence may need to:

- Find a teacher working in AI.
- Join an appropriate technical group.
- Find students with similar interests.
- Complete technical tasks.
- Receive feedback from teachers.
- Find research opportunities.
- Find project or hackathon teammates.
- Search for internships or jobs.
- Determine which skills are missing for a desired career.

These activities are highly interconnected, but traditional university platforms generally treat them as separate systems.

To address this problem, T2S is proposed as an AI-powered multi-agent intelligent platform that connects students, teachers, and alumni through a unified ecosystem for skill development, mentorship, collaboration, research, hackathons, knowledge sharing, and career discovery.

The system will use a multi-agent AI architecture in which specialized AI agents can perform different tasks while an AI orchestrator coordinates their interactions.

2. Problem Statement

University students face several challenges in developing technical and professional skills.

Major problems include:

1. Difficulty finding suitable teachers or mentors based on interests.
2. Lack of structured teacher-led technical groups.
3. Difficulty tracking practical skill development.
4. Limited mechanisms for teachers to assign and evaluate technical tasks.
5. Difficulty finding suitable teammates for projects and hackathons.
6. Research opportunities are often difficult to discover.
7. Learning resources are scattered across multiple platforms.
8. Students may not know which skills are required for their desired careers.
9. Internship and job opportunities are distributed across different sources.
10. Students have limited personalized guidance based on their actual activities and skills.

Existing platforms generally solve only one or two of these problems.

Therefore, T2S aims to create a unified intelligent platform where student activities, skills, mentorship, collaboration, research, and career opportunities are connected.

3. Proposed Solution

T2S will provide a centralized platform where students, teachers, and alumni can interact.

The platform will contain several interconnected modules:

- Teacher Mentorship
- Technical Rooms/Groups
- Task and Problem Management
- Student Skill Profiles
- Student Progress Tracking
- Project Collaboration
- Hackathon Team Formation
- Research Collaboration
- Knowledge Hub
- Job and Internship Opportunities
- Alumni Opportunities
- AI Assistant
- Multi-Agent AI System
- Personalized Recommendations

The core concept is:

$$\text{Student Activities} \rightarrow \text{Skill Analysis} \rightarrow \text{AI Recommendations} \rightarrow \text{Tasks/Teams/Research/Jobs} \rightarrow \text{New Activities} \rightarrow \text{Updated Skill Profile}$$

This creates a continuously evolving student development ecosystem.

4. Objectives

4.1 General Objective

The main objective of this project is to develop an AI-powered multi-agent platform that connects students, teachers, and alumni while supporting technical skill development, mentorship, collaboration, research, and career discovery.

4.2 Specific Objectives

1. To develop an integrated platform for student skill development and mentorship that enables teachers to create technical learning groups, assign and evaluate tasks, provide feedback, and monitor students' progress through dynamic skill profiles.
2. To implement an AI-powered multi-agent system for personalized learning, collaboration, and career development that analyzes student skills and interests to recommend relevant learning resources, research opportunities, project and hackathon teammates, internships, and job opportunities.
3. To establish a unified collaborative ecosystem connecting students, teachers, and alumni through knowledge sharing, research collaboration, career networking, and an AI assistant, while ensuring data privacy, responsible AI usage, and human oversight of important academic and career-related decisions.

5. Target Users

The system will primarily support three types of users.

5.1 Students

Students will be able to:

- Create profiles.
- Select technical interests.
- Join teacher-led rooms.
- Complete assigned tasks.
- Submit solutions.
- Receive marks and feedback.
- Track their progress.
- Build skill profiles.
- Find teammates.
- Join hackathons.
- Discover research opportunities.
- Search learning resources.
- Discover jobs and internships.
- Ask the AI assistant questions.

5.2 Teachers

Teachers will be able to:

- Create technical rooms.
- Accept students.
- Assign tasks.
- Set deadlines.
- Review submissions.
- Provide marks.
- Give feedback.
- Monitor student progress.
- Recommend resources.

- Identify students for research/project collaboration.

5.3 Alumni

Alumni will be able to:

- Share jobs.
- Share internships.
- Post project opportunities.
- Share industry knowledge.
- Connect with students.
- Recommend skills or learning paths.

6. Core System Modules

6.1 Authentication and User Management

The system will support role-based authentication.

Roles:

- Student
- Teacher
- Alumni
- Administrator

Users will have different permissions depending on their roles.

7. Teacher-Led Technical Rooms

Teachers will be able to create technical rooms based on different fields.

Example rooms:

- Artificial Intelligence
- Machine Learning
- Data Science
- Competitive Programming
- Web Development
- Cybersecurity
- Software Engineering
- Research

Students can browse available rooms and request to join them.

Teachers can approve or reject requests.

8. Task and Problem Management

Teachers can create technical tasks for students.

A task may contain:

- Title
- Description
- Difficulty
- Required skills
- Deadline
- Resources
- Submission requirements
- Maximum marks

Students can submit their solutions through the platform.

Teachers can:

- Review submissions
- Give marks
- Provide feedback
- Track completion
- Monitor student progress

9. Student Skill Profile

One of the major components of T2S will be the Student Skill Profile.

Instead of maintaining only a traditional user profile, the system will maintain a dynamic representation of the student's activities and skills.

The profile may contain:

Technical Skills

- Python
- JavaScript
- React
- FastAPI
- Machine Learning
- Deep Learning
- SQL
- Data Science
- NLP

Activities

- Completed tasks
- Projects
- Research
- Hackathons
- Competitions
- Teacher evaluations

Achievements

- Completed projects
- Research publications

- Hackathon participation
- Certificates
- Awards

The profile will evolve as the student performs activities on the platform.

10. Student Progress and Performance

The platform will provide interactive progress visualization.

For example:

Skill	Current Level	Completed Tasks	Teacher Feedback
Python	Advanced	18	Strong
Machine Learning	Intermediate	10	Good
React	Advanced	15	Strong
SQL	Intermediate	7	Needs Improvement

Teachers can use this information to understand student progress.

Students can use it to identify areas requiring improvement.

The performance table should be treated as a mentoring and progress indicator, not an official academic ranking.

11. AI-Powered Multi-Agent Architecture

The central technical component of T2S will be a Multi-Agent AI System.

Instead of using one general AI component for every task, the system will use specialized agents.

Proposed architecture:

$$\text{User} \rightarrow \text{AI Orchestrator} \rightarrow \text{Specialized Agent} \rightarrow \text{Tools/Data} \rightarrow \text{Result} \rightarrow \text{User}$$

12. AI Orchestrator

The AI Orchestrator will act as the central coordinator.

Its responsibilities include:

1. Understanding the user's request.
2. Identifying the appropriate agent.
3. Providing necessary context.
4. Calling authorized tools.
5. Combining results.
6. Returning the final response.

For example:

A student asks:

"I want to find an AI project team. I know Python and ML but I need someone good at frontend."

The orchestrator may identify that the request requires the Team Matching Agent.

The agent can search student profiles and identify candidates with complementary frontend skills.

13. Specialized AI Agents

13.1 Profile and Skill Agent

Responsibilities:

- Analyze student activities.
- Identify technical skills.
- Detect potential skill gaps.
- Update skill-related information.
- Generate skill development recommendations.

13.2 Task Agent

Responsibilities:

- Recommend relevant tasks.
- Analyze student skill levels.
- Suggest appropriate difficulty.
- Recommend learning activities.

Example:

"You have completed beginner ML tasks. The next suitable task is a classification project using real-world data."

13.3 Team Matching Agent

The Team Matching Agent will help students find appropriate teammates.

Matching will consider complementary skills rather than simply identical skills.

Example:

A project requires:

- Machine Learning
- Backend
- Frontend
- UI/UX

The agent can recommend a team containing students with complementary abilities.

13.4 Career Agent

The Career Agent will support career discovery.

It can:

- Find relevant jobs/internships from permitted sources.
- Normalize job information.
- Extract required skills.
- Match opportunities with student profiles.
- Identify skill gaps.

For example:

Student skills:
Python + ML + SQL

Job requirements:
Python + ML + SQL + Docker

The system can identify Docker as a potential skill gap.

13.5 Research Agent

The Research Agent will assist with research-related discovery.

It can help users find:

- Research topics
- Research papers
- Relevant resources
- Research opportunities
- Potential collaborators
- Teachers working in related areas

Academic sources should be accessed through permitted APIs, public metadata, or authorized resources.

13.6 Knowledge Agent

The Knowledge Agent will support the Knowledge Hub.

It can use Retrieval-Augmented Generation (RAG) over authorized resources.

Resources may include:

- Books
- Research papers
- Notes
- Tutorials
- Documentation
- Datasets
- Learning resources
- GitHub repositories

The agent should provide answers based on retrieved resources rather than relying only on the language model's internal knowledge.

14. AI Assistant

T2S will provide an interactive AI assistant.

Example questions:

“Which AI rooms can I join?”

“What tasks should I complete next?”

“Find students who can help me with frontend.”

“What skills do I need for an ML Engineer internship?”

“Find research opportunities related to computer vision.”

“Find suitable hackathon teammates.”

The assistant will use platform tools and specialized agents to provide context-aware responses.

15. AI Tool System

The AI agents should not directly access the database without controlled interfaces.

Instead, the system can provide authorized tools such as:

```
search_jobs()
search_students()
search_tasks()
search_teams()
search_rooms()
search_library()
search_research()
get_student_profile()
get_student_progress()
match_team_members()
```

This provides better control over what the AI can access.

16. Job and Internship Module

The Career Hub will provide job and internship opportunities.

The Job Collection Agent can collect opportunities through:

- Public APIs
- RSS feeds
- Public feeds
- Approved integrations
- Sources that permit automated collection

The system should not bypass:

- Authentication
- CAPTCHAs
- Access controls

- Robots restrictions
- Website terms

Collected opportunities can be:

1. Retrieved
2. Normalized
3. Deduplicated
4. Classified
5. Skill requirements extracted
6. Matched with students

17. Hackathon and Project Team Formation

The Hackathon Hub will allow students to create opportunities.

For example:

Hackathon

Hackathon: AI Innovation Challenge

Current Team:

- Student A — Machine Learning
- Student B — Backend

Required Skills:

- Frontend
- UI/UX

Other students can request to join.

The Team Matching Agent can recommend suitable candidates based on:

- Skills
- Interests
- Previous projects
- Availability
- Required roles
- Complementary abilities

The team leader will make the final decision.

18. Research Hub

The Research Hub will allow students to discover research opportunities.

Features may include:

- Research topics
- Research papers
- Research resources
- Teacher research interests

- Student research interests
- Research collaboration
- Research project opportunities

The AI Research Agent can help students identify relevant resources and potential collaborators.

19. Knowledge Hub

The Knowledge Hub will provide a centralized learning-resource section.

Possible resources:

- Books
- Papers
- Tutorials
- Documentation
- Notes
- Datasets
- GitHub repositories

Users can search the resources using normal keyword search or semantic search.

The Knowledge Agent can use RAG to answer questions based on authorized resources.

Copyright restrictions must be respected.

The platform should preferably store:

- Metadata
- Links
- Openly licensed resources
- Authorized content

rather than unauthorized copies of copyrighted material.

20. Alumni Career Network

The Alumni module will connect current students with graduates.

Alumni can:

- Post jobs.
- Post internships.
- Share projects.
- Share industry advice.
- Recommend skills.
- Connect with students.

This creates a connection between:

$$\text{Student} \rightarrow \text{University} \rightarrow \text{Alumni} \rightarrow \text{Industry}$$

21. Recommendation System

The platform will provide personalized recommendations.

Possible recommendation categories:

Task Recommendation
"Complete these three tasks to improve your ML skills."

Resource Recommendation
"These tutorials are relevant to your current skill level."

Team Recommendation
"These students have complementary skills for your project."

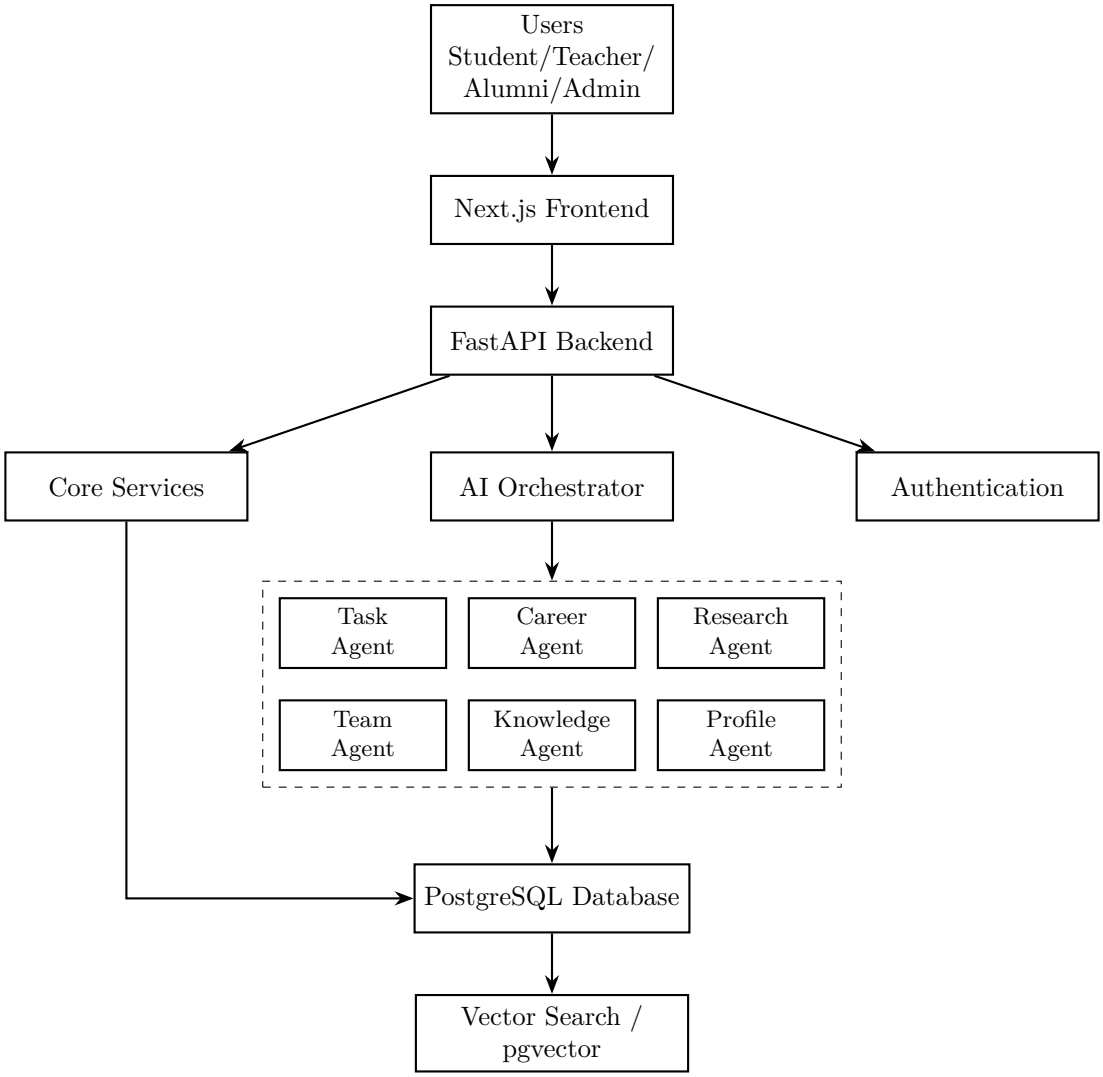
Research Recommendation
"These research topics match your interests."

Career Recommendation
"These internships match approximately 80% of your current skills."

Recommendations should be treated as decision-support information rather than guaranteed outcomes.

22. System Architecture

The proposed architecture is:



23. Technology Stack

Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS

Backend

- Python
- FastAPI
- Pydantic

Database

- PostgreSQL

Vector Search

- pgvector

AI

- LLM API or suitable open-source LLM
- Embedding model
- RAG
- Multi-agent orchestration

Machine Learning

- Scikit-learn
- NumPy
- Pandas

Authentication

- JWT or secure session-based authentication

Testing

- Postman
- Pytest
- Frontend testing tools where appropriate

Development

- Git
- GitHub
- Docker

24. Database Entities

The major database entities may include:

User	Task	Job
StudentProfile	Submission	Internship
TeacherProfile	Evaluation	ResearchOpportunity
AlumniProfile	Progress	Resource
Skill	Project	Notification
Interest	Team	AIInteraction
Room	TeamMember	Recommendation
RoomMembership	Hackathon	

Relationships between these entities will allow the system to build a connected student ecosystem.

25. Functional Requirements

FR-01 User Registration
The system shall allow users to create accounts according to their roles.

FR-02 Authentication
The system shall provide secure authentication.

FR-03 Profile Management
Users shall be able to create and update their profiles.

FR-04 Room Management
Teachers shall be able to create and manage technical rooms.

FR-05 Room Membership
Students shall be able to request to join rooms.

FR-06 Task Management
Teachers shall be able to create and manage tasks.

FR-07 Submission
Students shall be able to submit completed tasks.

FR-08 Evaluation
Teachers shall be able to evaluate submissions.

FR-09 Progress Tracking
The system shall track student activities and progress.

FR-10 Skill Profile
The system shall maintain student skill information.

FR-11 Collaboration
Students shall be able to create and join project teams.

FR-12 Hackathon
Students shall be able to post and join hackathon teams.

FR-13 Research
Users shall be able to discover research opportunities and resources.

FR-14 Knowledge Hub
Users shall be able to search learning resources.

FR-15 Career Hub
Users shall be able to search job and internship opportunities.

FR-16 Alumni
Alumni shall be able to post career opportunities.

FR-17 AI Assistant
Users shall be able to interact with the AI assistant.

26. AI Requirements

AI-01 Multi-Agent Architecture
The system shall implement multiple specialized AI agents.

AI-02 Agent Orchestration
The system shall contain an orchestrator capable of routing requests to appropriate agents.

AI-03 Tool Usage
Agents shall use authorized platform tools to retrieve relevant information.

AI-04 Personalized Recommendation
The system shall generate recommendations based on available student information.

AI-05 RAG

The Knowledge Agent shall retrieve relevant authorized resources before generating answers where applicable.

AI-06 Skill Analysis
The system shall analyze student activities and skills.

AI-07 Human-in-the-Loop
Important academic, team, and career decisions shall remain under human control.

27. Non-Functional Requirements

Performance
The system should provide reasonable response times for normal user operations.

Security
The platform should protect:

- User credentials
- Personal information
- Academic information
- Private submissions
- Teacher feedback

Scalability
The architecture should allow additional departments, rooms, users, and AI agents to be added later.

Maintainability
The backend should use modular architecture.

Reliability
The system should handle failures gracefully.

Usability
The user interface should be understandable for students, teachers, and alumni.

28. AI Safety and Ethics

Since T2S will process educational and career-related information, responsible AI practices are required.

The system should:

1. Protect user privacy.
2. Avoid exposing private student information.
3. Clearly distinguish AI recommendations from human decisions.
4. Avoid presenting recommendations as guaranteed outcomes.
5. Allow teachers to review important decisions.
6. Avoid discriminatory recommendations.
7. Respect copyright.
8. Respect external website terms.
9. Avoid unauthorized data collection.
10. Maintain appropriate access controls.

AI should assist teachers and students rather than replace them.

29. Evaluation Strategy

The system will be evaluated from both software and AI perspectives.

29.1 Software Evaluation

Testing will include:

- Unit testing
- API testing
- Integration testing
- Authentication testing
- Database testing
- UI testing
- Security testing

29.2 AI Evaluation

Possible evaluation criteria include:

Recommendation Quality
Evaluate whether recommended:

- Tasks
- Resources
- Students
- Teams
- Jobs
- Research opportunities

are relevant.

Retrieval Quality
Evaluate whether the RAG system retrieves relevant resources.

Agent Routing
Measure whether user requests are correctly routed to appropriate agents.

Human Evaluation
Teachers/students can evaluate AI responses for:

- Relevance
- Accuracy
- Helpfulness
- Groundedness

The project should avoid claiming production-level AI accuracy if the evaluation dataset is small.

30. Project Scope

In Scope

- Student management
- Teacher management
- Alumni management
- Technical rooms
- Tasks
- Submissions
- Teacher evaluation
- Progress tracking
- Skill profiles
- AI assistant
- Multi-agent AI
- Team matching
- Hackathon collaboration
- Knowledge Hub
- Research Hub
- Job/internship discovery
- Alumni opportunities

Out of Scope

The initial version will not attempt to:

- Replace university academic grading systems.
- Replace professional recruitment systems.
- Guarantee employment.
- Automatically make final hiring decisions.
- Automatically make academic decisions.
- Bypass restricted job websites.
- Host unauthorized copyrighted books.
- Provide unrestricted autonomous AI actions.

31. Project Timeline

The proposed development period is 16 weeks.

Week	Activity
1–2	Requirement Analysis
2–3	UI/UX Design
3–4	Database Design
4–5	Authentication
5–7	Backend Development
6–9	Frontend Development
8–10	Task & Progress System
9–11	Collaboration & Team System
10–11	AI Foundation
11–13	Multi-Agent AI
12–14	Knowledge & Research Hub
12–14	Career/Job Module
14–15	Full System Integration
15–16	Testing & Optimization
16	Documentation & Final Presentation

32. Team Member Responsibilities

For a Three-member team:

Team Member Responsibilities

Member 1 Frontend Development and UI/UX Design

Responsibilities:

- Develop the frontend using Next.js, React, TypeScript, and Tailwind CSS.
- Design and implement student, teacher, and alumni dashboards.
- Develop user profile, technical room, task, and progress-tracking interfaces.
- Build the Knowledge Hub, Research Hub, and Career Hub interfaces.
- Integrate frontend components with backend APIs.
- Ensure responsive design and a user-friendly experience.

Member 2 Backend Development and Database Management

Responsibilities:

- Develop backend services using Python and FastAPI.
- Design and manage the PostgreSQL database.
- Implement authentication, authorization, and role-based access control.
- Develop REST APIs for profiles, technical rooms, tasks, submissions, and evaluations.
- Implement project collaboration, hackathon management, and alumni opportunity services.
- Develop notification, data validation, and security mechanisms.
- Integrate backend services with the frontend and AI modules.

Member 3 AI/ML Development and Multi-Agent System

Responsibilities:

- Design and implement the AI orchestrator and specialized AI agents.
- Develop the Profile and Skill Agent, Task Agent, Team Matching Agent, Career Agent, Research Agent, and Knowledge Agent.
- Implement Retrieval-Augmented Generation (RAG) and embedding-based semantic search.
- Develop personalized task, resource, team, research, and career recommendations.
- Implement skill-gap analysis and student-opportunity matching.
- Integrate AI agents with authorized backend tools and platform data.
- Evaluate AI responses, recommendation quality, and agent-routing accuracy.
- Ensure responsible AI usage, data privacy, and human oversight.

Collaborative Responsibilities

All three members will collaborate on system integration, testing, debugging, documentation, security review, and final project presentation.

33. Expected Outcome

At the completion of the project, T2S is expected to provide a unified platform where:

Students can

- Find mentors.
- Join technical communities.
- Develop practical skills.
- Complete tasks.
- Track progress.
- Find collaborators.
- Join hackathons.
- Discover research opportunities.
- Access learning resources.
- Discover jobs and internships.

Teachers can

- Create technical communities.
- Assign tasks.
- Evaluate students.
- Monitor progress.
- Provide feedback.
- Identify promising students for projects and research.

Alumni can

- Share jobs.
- Share internships.
- Post projects.
- Connect with students.
- Provide industry guidance.

AI can

- Understand user requests.

- Route requests to specialized agents.
- Search authorized platform data.
- Analyze student skills.
- Recommend tasks and resources.
- Match students with teams.
- Match students with opportunities.
- Support research and career discovery.

34. Future Work

Future versions of T2S may include:

- University-wide deployment.
- Mobile application.
- Advanced skill graphs.
- More sophisticated recommendation models.
- Integration with university LMS systems.
- Integration with GitHub.
- Advanced research collaboration.
- Industry partnerships.
- More external career integrations.
- Advanced analytics for teachers.
- Multilingual AI support.
- More specialized AI agents.

35. Expected Impact

T2S aims to create a more connected university technical ecosystem.

Instead of students independently searching for mentors, resources, teammates, research opportunities, and jobs, the platform connects these activities.

The expected ecosystem is:

Student
↓
Skills & Interests
↓
Learning & Tasks
↓
Teacher Feedback
↓
Skill Development
↓
Projects / Research / Hackathons
↓
Career Opportunities
↓
Industry / Alumni

This creates a continuous development cycle for students.

36. Conclusion

T2S proposes an AI-powered multi-agent platform designed to address several challenges faced by university students in technical skill development, mentorship, collaboration, research, and career discovery.

The system combines conventional web application functionality with modern AI technologies including LLMs, Retrieval-Augmented Generation, embeddings, recommendation systems, tool-based AI, and multi-agent orchestration.

The main contribution of T2S is not simply providing an AI chatbot. Instead, AI will be integrated with the platform's real data and services so that specialized agents can perform meaningful tasks such as skill analysis, task recommendation, team matching, research discovery, knowledge retrieval, and career matching.

The system will maintain human oversight for important academic and career-related decisions.

Therefore, T2S can serve as a practical and technically meaningful university project demonstrating the integration of full-stack software engineering, databases, AI, machine learning, natural language processing, recommendation systems, and multi-agent intelligent systems into a single platform.